

## ASYMPTOTIC EFFICIENCY OF A CLASS OF RANK TESTS FOR MULTIPLE LINEAR REGRESSION

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### SUMMARY

A rank score test statistic is constructed for the general linear regression parameters. The statistic is shown to be the same as some well-known nonparametric statistics for some familiar distribution functions. Under certain null and alternative hypotheses, it is also shown to have limiting central and noncentral chi-square distributions respectively. Finally a measure of efficiency due to Pitman is used to find the asymptotic efficiency of the new statistic relative to its classical counterpart.

### 1. Introduction

Hájek (1962) constructed an asymptotically most powerful rank score statistic for testing hypotheses in certain linear regression models. Following this, Adichie (1967) developed a sign-rank statistic for tests in a two-parameter linear regression model. Jogdeo (1964), however, extended Hájek's procedure by testing the hypothesis:  $\beta_1 = \beta_2 = \dots = \beta_p = 0$  in the regression model:  $Y_{nj} = \beta_0 + \beta_1 x_{nj1} + \dots + \beta_p x_{njp} + \sigma \epsilon_j$ . Like Hajek, Jogdeo regarded  $\beta_0$  as a nuisance parameter and assumed that the distribution of the random observations,  $Y_{nj}$ , is known.

The present investigation generalizes Adichie's results by using a statistical model similar to that of Jogdeo in obtaining a sign-rank statistic for the hypothesis:  $\beta_0 = \beta_1 = \beta_2 + \dots + \beta_p = 0$ .

### 2. Assumptions and Notations

Let  $(Y_{n1}, \dots, Y_{nn})$  be a sequence of random vectors. The  $Y_{nj}$ ,  $j=1, \dots, n$ , are independent with distributions:

$$(2.1) P_{\beta}(Y_{nj} \leq y) = F(y - \sum_k \beta_k x_{nj k})$$

where  $x_{nj0} \equiv 1$  for all  $j$ , the summation over  $k$  goes from 0 to  $p$ , and  $P_{\beta}$  denotes that the probability is being computed for the parameter values  $\beta = (\beta_0, \beta_1, \dots, \beta_p)$ .

The  $x_{nj k}$  are regression constants depending on  $n$ . In certain cases this dependence on  $n$  of the  $x$ 's and  $Y$ 's will be suppressed for convenience provided no confusion arises.

The problem here is to construct rank score tests for the hypothesis:

$$(2.2) H_0 : \beta=0, \beta = (\beta_0, \beta_1, \dots, \beta_p).$$

The form of  $F$  is not known but it is assumed that  $F \in \mathcal{F}$ , where the class  $\mathcal{F}$  is defined by Adichie (1967). We shall also assume that the constants  $x_{jk}$  satisfy the following conditions:

$$(2.3) \lim \{ \max_j (\sum_k x_{jk})^2 \} / \{ n^{-1} \sum_j (\sum_k x_{jk})^2 \} = 0$$

$$(2.4) 0 < \lim n^{-1} \sum_j (\sum_k x_{jk})^2 < \infty$$

$$(2.5) |\lim n^{-1} \sum_j \sum_k x_{jk}| < \infty$$

where  $|\cdot|$  denotes the absolute value, the summations over  $k$  and  $j$  go from 0 to  $p$  and 1 to  $n$  respectively, and the limits are taken as  $n \rightarrow \infty$ . The notation  $L(X_n/P) \rightarrow N(a, b^2)$  together with the class of functions  $\psi(u)$  and  $\phi(u)$  defined by Adichie (1967) will be used in the sequel.

### 3. A nonparametric test statistic

Let  $R_j$  be the rank of  $|Y_j|$  in the sequence of absolute values  $|Y_1|, \dots, |Y_n|$  of the  $n$  observations. Define a set of  $(p+1)$  statistics by

$$(3.1) T_{nk} = n^{-1} \sum_j x_{jk} \psi_n(R_j/(n+1)) \text{ sign } Y_j, k=0, 1, \dots, p, \text{ where}$$

$$(3.2) \psi_n(u) = \psi(j/(n+1)), (j-1)/n < u \leq j/n \text{ and}$$

$$(3.3) \lim \int_0^1 \{ \psi_n(u) - \psi(u) \}^2 du = 0$$

by [4]. Let a symmetric  $(p+1) \times (p+1)$  matrix  $\|\lambda_{ik}\|_n$  be given by

$$(3.4) \quad \begin{cases} \lambda_{n,kk} = \lambda_{nk}^2 = n^{-1} \sum_j x_{jk}^2 \int_0^1 \psi^2(u) du \text{ and} \\ \lambda_{n,ik} = \lambda_{n,ki} = n^{-1} \sum_j x_{ji} x_{jk} \int_0^1 \psi^2(u) du \end{cases}$$

where  $i, k = 0, 1, \dots, p$ . Now let

$$(3.5) M_n = n T_n \|\lambda_{ik}\|_n^{-1} T_n'$$

where  $T_n = (T_{n0}, T_{n1}, \dots, T_{np})$ ,  $T'$  is the transpose of  $T$  and  $||\lambda_{ik}||_n^{-1}$  is the inverse of  $||\lambda_{ik}||_n$ .  $M_n$  is therefore the proposed test statistic for  $H_0$ .

To every  $G \in \mathcal{F}$  corresponds one  $M_n$  through  $\psi(u)$  and  $\psi_n(u)$ . In particular, (cf. Adichie (1967)), if  $G$  is the logistic distribution function, the corresponding  $M_n$ -statistic is of the Wilcoxon type. If  $G$  is the standard normal distribution function,  $M_n$  is of the Van der Waerden type.

#### 4. Limiting distribution of $M_n$ under $H_0$

Since under  $H_0$  the joint distribution of  $T_{nk}$  ( $k=0,1,\dots,p$ ) is independent of  $F$ , but rather depends on  $G$  through  $\psi$ , then the limiting null distribution of  $M_n$  is given by the following theorem.

##### Theorem 4.1

*Under the assumptions of section 2*

$$\lim P_0(M_n \leq y) = P(\chi_{p+1}^2 \leq y)$$

where  $\chi_{p+1}^2$  is the central chi-square random variable with  $(p+1)$  degrees of freedom and  $P_0$  denotes that the probability is being computed under  $H_0$ .

##### Proof

It is sufficient to prove that  $L(n^{1/2}(T_{n0}, T_{n1}, \dots, T_{np})/P_0)$  tends to a  $(p+1)$ -variate normal distribution with covariance matrix  $||\lambda_{ik}|| = \lim ||\lambda_{ik}||_n$ . The method of proof used here, which is similar to that in Adichie (1967) or Hajek (1962) replaces  $T_{nk}$ ,  $k=0,1,\dots,p$ , by sums of independent random variables and applies central limit theorems.

To this effect we introduce the following two sets of statistics:

$$(4.1) S_{nk}^0 = n^{-1} \sum_j x_{jk} \psi_n(U_{nj}) \text{ Sign } Y_{nj} \text{ and}$$

$$(4.2) S_{nk} = n^{-1} \sum_j x_{jk} \psi(U_{nj}) \text{ Sign } Y_{nj}, k=0,1,\dots,p$$

where  $U_{nj} = G^*(|Y_{nj}|)$ , ( $j=1,\dots,n$ ), are independent random variables uniformly and identically distributed on  $(0,1)$ ,  $G^*(y) = P_0(|Y_{nj}| \leq y) = 2G(y)-1$  for  $y>0$ , and  $x_{j0} \equiv 1$  for all  $j=1$  to  $n$ . It is easily checked that for all  $k=0,1,\dots,p$

$$(4.3) E_0(n^{1/2} S_{nk}) = 0 \text{ and}$$

$$(4.4) \text{Var}_0(n^{1/2} S_{nk}) = n^{-1} \sum_j x_{jk}^2 E_0 \psi^2(U_{nj}) = n^{-1} \sum_j x_{jk}^2 \int_0^1 \psi^2(u) du = \lambda_{nk}^2$$

where  $E_0$  and  $\text{Var}_0$  denote that both the expectation and the variance are taken under  $H_0$ , and where  $\lambda_{nk}^2$  is defined in (3.4). Using the procedure in Adichie (1967) it can be proved that for  $k=0,1,\dots,p$

$$(4.5) \quad n^{1/2}(T_{nk} - S_{nk}^0) \rightarrow 0 \text{ in } P_0 - \text{probability and}$$

$$(4.6) \quad n^{1/2}(S_{nk}^0 - S_{nk}) \rightarrow 0 \text{ in } P_0 - \text{probability}$$

implying that for  $k=0,1,\dots,p$

$$(4.7) \quad L(n^{1/2}T_{nk}/P_0) \rightarrow L(n^{1/2}S_{nk}/P_0).$$

where  $L(T/P)$  has already been defined in section 2. Since  $n^{1/2}S_{nk}$  ( $k=0,1,\dots,p$ ) are sums of independent identically distributed random variables, the general central limit theorem of Loève (1963), Theorem B, p. 280) applies and because of (4.3), (4.4) and (4.7) we obtain for  $k=0,1,\dots,p$  that

$$(4.8) \quad L(n^{1/2}T_{nk}/P_0) \rightarrow N(0, \lambda_k^2)$$

where  $\lambda_k^2 = \lim \lambda_{nk}^2$  defined in (4.4). Now to prove the joint asymptotic normality of  $(n^{1/2}(T_{n0}, T_{n1}, \dots, T_{np})/P_0)$  it suffices because of (4.7) and a well-known theorem (Cramer (1954), p. 299) to obtain the joint asymptotic normality of  $(n^{1/2}(S_{n0}, S_{n1}, \dots, S_{np})/P_0)$ . This follows at once from Adichie (1967). Hence  $L(n^{1/2}(T_{n0}, T_{n1}, \dots, T_{np})/P_0)$  tends to a  $(p+1)$ -variate normal distribution with means zero and covariance matrix  $||\lambda_{ik}||$ . The assertion of the theorem follows immediately from the usual transformation contained, say, in Sverdrup (1952) and from the application of a well-known theorem on the distribution of quadratic forms.

## 5. Limiting distribution of $M_n$ under near alternatives

To determine the efficiency of  $M_n$ , it is necessary to find its distribution under a sequence of alternatives,  $H_n$ , which tends to  $H_0$  for large  $n$ . The sequence of alternatives used here is of the form:

$$(5.1) \quad H_n : \beta_{nk} = b_k n^{-1/2}, b_k \neq 0 \text{ for all } k=0,1,\dots,p.$$

For this purpose we shall employ the method based on LeCam's contiguity lemma (cf. Hájek (1962) and LeCam (1960)). The set-up under which this contiguity principle is applicable is discussed in Adichie (1967) and Hájek (1962). The following notations also defined by Adichie (1967) will be used. Let

$$(5.2) \quad P_0 = \pi_j P_{0j} \text{ and } P_n = \pi_j P_{nj}$$

be the joint probability distributions of  $(Y_{n1}, \dots, Y_{nn})$  under  $H_0$  and  $H_n$  respectively. Let

$$(5.3) \quad r_{nj} = dP_{nj}(Y_{nj} \leq y) / dP_{0j}(Y_{nj} \leq y) = P_{nj} / P_{0j}, \quad P_{0j} > 0.$$

where  $p_{0j}$  and  $p_{nj}$  are the densities corresponding to  $P_{0j}$  and  $P_{nj}$  respectively. Also let

$$(5.4) \quad W_n = 2 \sum_j (r_{nj}^{1/2} - 1).$$

Using the above notations we shall prove the following Lemma:

*Lemma 5.1*

*Under the regularity conditions on  $F$  given in Adichie (1967) and those on  $x_{jk}$  given in (2.3) to (2.5) if*

$$(i) \quad \lim \max_j P_0(|r_{nj} - 1| > \epsilon) = 0 \text{ for every } \epsilon > 0 \text{ and}$$

$$(ii) \quad L(W_n / P_0) \rightarrow N(-\frac{1}{4} \sigma^2, \sigma^2)$$

where  $\sigma^2$  is to be defined, then the distributions  $P_n$  are contiguous to  $P_0$ .

*Proof*

The proof of (i) follows directly from Adichie (1967). To prove (ii), define

$$(5.5) \quad S_n^* = n^{-1} \sum_j \sum_k b_k x_{jk} \{f'(Y_{nj}) / f(Y_{nj})\}, \quad x_{j0} \equiv 1 \text{ for all } j.$$

where  $f'(y) = (d/dy)f(y)$ . It can be easily checked that

$$(5.6) \quad E_0(n^{1/2} S_n^*) = 0 \text{ and}$$

$$(5.7) \quad \text{Var}_0(n^{1/2} S_n^*) = n^{-1} \sum_j (\sum_k b_k x_{jk})^2 \int_{-\infty}^{\infty} \{f'(y) / f(y)\}^2 f(y) dy$$

$$= n^{-1} \sum_j (\sum_k b_k x_{jk})^2 \int_0^1 \phi^2(u) du = \sigma_n^2$$

where  $E_0$  and  $\text{Var}_0$  denote that both the expectation and the variance are taken under  $H_0$  and  $\phi(u)$  is defined by Adichie (1967). Using (2.1) and (2.2) in (5.4),  $W_n$  may be

written in this form

$$(5.8) W_n = 2 \sum_j \{s(Y_{nj} - h_{nj})/s(Y_{nj}) - 1\}$$

where  $s(y) = f^{1/2}(y)$  and  $h_{nj} = n^{-1/2} \sum_k b_k x_{ijk}$  and it is assumed that  $h_{nj} \neq 0$ . Now the derivation of the following results follows the same way as that given in Hájek and Sidak (1967, pp. 211-213).

$$(5.9) E_0 W_n \rightarrow -\frac{1}{4} \sigma^2 \text{ and}$$

$$(5.10) E_0(W_n - E_0 W_n - n^{1/2} S_n^*)^2 \rightarrow 0 \text{ as } n \rightarrow \infty, \text{ where}$$

$$(5.11) \sigma^2 = \lim \sigma_n^2 \text{ defined in (5.7).}$$

Since  $n^{1/2} S_n^*$  is the sum of independent identically distributed random variables with mean zero and variance  $\sigma_n^2$ , by the central limit theorem of (Loève (1963), Theorem B, p. 280) we have

$$(5.12) L(n^{1/2} S_n^*/P_0) \rightarrow N(0, \sigma^2)$$

It then follows from (5.9), (5.10), (5.11) and (5.12) that

$$(5.13) L(W_n/P_0) \rightarrow N(-\frac{1}{4} \sigma^2, \sigma^2).$$

and this completes the proof of the Lemma.

The principle of contiguity will now be applied to obtain  $L(n^{1/2}(T_{n0}, T_{n1}, \dots, T_{np})/H_n)$ . Hence the following theorem:

#### Theorem 5.1

Under the assumptions of Lemma 5.1,  $L(n^{1/2}(T_{n0}, T_{n1}, \dots, T_{np})/H_n)$  tends to the  $(p+1)$ -variate normal distribution with mean vector  $(\mu_0, \mu_1, \dots, \mu_p)$  and covariance matrix  $\|\lambda_{jk}\|$ , where the means are defined by

$$(5.14) \mu_k = \lim n^{-1} \sum_j \sum_i b_i x_{ijk} \int_0^1 \psi(u) \phi(u) du$$

#### Proof

The proof follows the same pattern as in Adichie (1967). Firstly, we prove that  $L(n^{1/2} T_{nk}/P_n) \rightarrow N(\mu_k, \lambda_k^2)$   $k=0, 1, \dots, p$ , and then establish the joint asymptotic normality of  $(n^{1/2}(T_{n0}, T_{n1}, \dots, T_{np})/P_n)$ . The first follows if we show that  $L(n^{1/2} T_{nk}, W_n/P_0)$  and

hence  $L(n^{1/2}S_{nk}, n^{1/2}S_n^*/P_0)$ , because of (4.7) and (5.10), tend to bivariate normal distributions with certain correlation coefficients (to be specified). It is easily seen from (4.2) and (5.5) that for  $k=0, 1, \dots, p$

$$(5.15) \quad \text{cov}_0 n^{1/2}(S_{nk}, S_n^*) = n^{-1} \sum_j \sum_i b_i x_{jk} \int_0^1 \psi(u) \phi(u) du = c_{nk}$$

where the summation over  $i$  is from 0 to  $p$ . Since  $\lim c_{nk}$  exists because of (2.4), the bivariate central limit theorem of Cramér (1963, Theorem 21a, p. 114) applies showing that  $L(n^{1/2}S_{nk}, n^{1/2}S_n^*/P_0)$  ( $k=0, 1, \dots, p$ ) tend to the bivariate normal distributions with correlation coefficients given by (cf. (4.4), (5.7) and (5.15)):

$$(5.16) \quad \rho_k = \lim c_{nk} / \lambda_{nk} \sigma_n.$$

Apply (3) of Lemma 4.2 of Hájek (1962) with  $\rho_k$  to obtain that  $L(n^{1/2}T_{nk}/P_n)$  tend to normal distributions with means  $\mu_k$  and variances  $\lambda_k^2$ . The proof of the asymptotic normality of  $(n^{1/2}(T_{n0}, T_{n1}, \dots, T_{np})/P_n)$  follows from Adichie (1967). This completes the proof of Theorem 5.1.

The limit distribution of  $M_n$  under  $H_n$  is then stated in the following theorem:

#### Theorem 5.2

Under the assumptions of Lemma 5.1 and Theorem 5.1,  $L(M_n/P_n) \rightarrow \chi_{p+1}^2(\Delta^2)$  where  $\chi_{p+1}^2(\Delta^2)$  denotes the noncentral chi-square distribution with  $(p+1)$  degrees of freedom and noncentrality parameter given by

$$(5.17) \quad \Delta^2 = \lim \{ (\sum_k b_k \bar{x}_{nk})^2 + n^{-1} \sum_j \sum_k b_k^2 (x_{jk} - \bar{x}_{nk})^2 \} \{ (\int_0^1 \psi(u) \phi(u) du)^2 / \int_0^1 \psi^2(u) du \}$$

where  $\bar{x}_{nk} = n^{-1} \sum_j x_{jk}$  and  $x_{j0} \equiv 1$  for all  $j$ .

#### Proof

The proof follows directly from that of theorem 5.1 and  $\Delta^2$  in (5.17) is obtained by a straightforward computation.

### 6. Asymptotic efficiency of $M_n$

We employ a measure of relative efficiency of two test statistics due to Pitman (cf. Noether (1954)). Moreover it is shown by Andrews (1954) and Hannan (1956) that, if, under the same sequence of alternatives, two test statistics have noncentral chi-square limit distributions with the same degrees of freedom, then their relative asymptotic efficiency is given by the ratio of their noncentrality parameters. Hence, to find the asymptotic efficiency of  $M_n$  relative to its classical counterpart, we only need to compute the noncentrality parameter of the latter.

The classical test statistic corresponding to  $M_n$  is (cf. Adichie (1967)) given by

$$(6.1) \quad M_n^* = n\beta_n^* \|\gamma_{ik}\|_n^{-1} \beta_n^{*'}, \beta_n^* = (\beta_{n0}^*, \beta_{n1}^*, \dots, \beta_{np}^*)$$

where  $\beta_{nk}^*$ ,  $k=0,1,\dots,p$ , are the least squares (or maximum likelihood) estimates of  $\beta_k$ , and  $\|\gamma_{ik}\|_n^{-1}$  is the inverse of  $\|\gamma_{ik}\|_n$  defined by

$$(6.2) \quad \|\gamma_{ik}\|_n = \|\tau_{ik}\|_n^{-1} \text{ where}$$

$$(6.3) \quad \tau_{ik} = n^{-1} \sum_j x_{ji} x_{jk} \quad (i,k=0,1,\dots,p).$$

If, as is usual in classical tests, the underlying distribution  $F$  is assumed to be the standard normal distribution function  $\Phi$ , then the estimates  $\beta_{nk}^*$ ,  $k=0,1,\dots,p$ , being linear functions of normal random variables are themselves normal random variables. Hence, under  $H_0$ ,  $M_n^*$  has a central chi-square distribution with  $(p+1)$  degrees of freedom and under any given alternatives of the form  $\beta_k = b_k$  ( $k=0,1,\dots,p$ ),  $M_n^*$  has a noncentral chi-square distribution with  $(p+1)$  degrees of freedom and a noncentrality parameter given by (cf. Lehmann (1959, p. 284)):

$$(6.4) \quad \Delta^{**2} = \lim \{n(\sum_k b_k \bar{x}_{nk})^2 + n^{-1} \sum_k b_k^2 (x_{jk} - \bar{x}_{nk})^2\}$$

where  $\bar{x}_{nk} = n^{-1} \sum_j x_{jk}$ . Hence from (5.17) and (6.5) the asymptotic efficiency of  $M_n$  relative to  $M_n^*$  becomes

$$(6.6) \quad e(M_n, M_n^*) = (\int_0^1 \psi(u) \phi(u) du)^2 / \int_0^1 \psi^2(u) du.$$

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